

LOCALIZED GL_2 BOST–CONNES SYSTEMS: THE EQUILIBRIUM RANGE AND THE AUTOMORPHIC OBSTRUCTION RANGE ARE DISJOINT, BY EXACTLY ONE

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ABSTRACT. We examine the proposal to characterize Langlands reciprocity for weight-two forms through the KMS structure of a localized GL_2 Bost–Connes system, and find a quantitative obstruction that closes it.

The Hilbert–Schmidt identity is correct: with $U_p = p^{-1/2}\rho_f(\text{Frob}_p)$ unitary by Deligne, $\|I_2 - U_p\|_{\text{HS}}^2 = 4(1 - \cos \theta_p) = 8\sin^2(\theta_p/2)$, and $\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$ for every cuspidal eigenform, by Jacquet–Shalika. But the L -function formula proposed for it is wrong: Ξ_β is *linear* in $\text{Tr } U_p$, so only the standard L -function appears,

$$\Xi_\beta(\pi_f, S) = 4 \log \zeta_S(\beta) - 2 \Re \log L_S(\beta, \pi_f) + O(1),$$

matching Theorem 4.1 of Paper VI of the series at $n = 2$; the term $-2\Re \log L_S(\beta, \text{Sym}^2 \pi_f)$ is spurious. Rankin–Selberg governs the second moment $\sum_p |\text{Tr } U_p|^2 p^{-s}$, which is Sato–Tate, not Ξ_β .

The decisive obstruction is a mismatch of temperature ranges, and it is exact. For any S , the automorphic obstruction can be nontrivial only where $\sum_{p \in S} p^{-\beta}$ diverges, since $\Xi_\beta \leq 8 \sum_{p \in S} p^{-\beta}$; write $\beta_c(S) \leq 1$ for that abscissa. The GL_2 partition function has local factor $(1 - p^{-\beta})^{-1}(1 - p^{1-\beta})^{-1}$ — the Solomon zeta of $M_2(\mathbb{Z}_p)$ — so it converges exactly for $\beta > 1 + \beta_c(S)$. Hence

$$\text{obstruction nontrivial} \iff \beta \leq \beta_c(S), \quad \text{Gibbs states exist} \iff \beta > 1 + \beta_c(S),$$

and the two regions are disjoint with a gap of width exactly 1, for every S and every automorphic π . For $S = P_{\mathbb{Q}}$: obstruction on $(0, 1]$, equilibrium on $(2, \infty)$. There is no temperature at which the GL_2 system has equilibrium states and the automorphic obstruction can break symmetry.

The shift by one is not accidental. The number of sublattices of index p in $\mathbb{Z}_p^2$ is $p+1$ while the norm is p , and by Theorem 1.2 of Paper XII this excess is universal: $d_p = (q^d - 1)/(q - 1) > q$. The same off-by-one of $\mathbb{P}^{d-1}$ that empties the boundary quotient there displaces the equilibrium range here.

Finally, the proposed algebra is not available as stated: $M_2(\mathbb{Z})^+$ has infinite unit group $SL_2(\mathbb{Z})$, so by the unit-group exclusion of Paper X (its Remark 1.5) its partition function diverges identically and one must pass to the Hecke quotient, which is not a semigroup crossed product.

1. THE ALGEBRA

The framework is that of [6]; Papers VI, X and XII of the series, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place.

Proposition 1.1 (The semigroup crossed product is unavailable). *Let $P = M_2(\mathbb{Z}) \cap GL_2^+(\mathbb{Q})$. Then $P^\times = SL_2(\mathbb{Z})$ — an element of P with inverse in P has positive integral determinant on both sides, hence determinant 1 — is infinite and $|\det| = 1$ on it, so $\sum_{g \in P} |\det g|^{-\beta} \geq |P^\times| = \infty$ for every β . Hence $C(Y) \rtimes P$ carries no Gibbs state at any temperature, and the construction must be replaced by the Hecke-algebra quotient of Connes–Marcolli [1], in which one sums over $GL_2(\mathbb{Z})$ -orbits.*

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XXI-GL2/](#).

Proof. This is the unit-group exclusion recorded in Paper X (its Remark 1.5): a multiplicative norm is 1 on units, so an infinite unit group makes ζ_P diverge identically. $\square$

Proposition 1.2 (The Solomon zeta). *Summing over $GL_2(\mathbb{Z})$ -orbits, the local factor at p is the Solomon zeta of $M_2(\mathbb{Z}_p)$ [5],*

$$\zeta_{GL_2,p}(\beta) = (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1},$$

so for a set S of primes

$$(1.1) \quad \zeta_{GL_2,S}(\beta) = \prod_{p \in S} (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1},$$

which converges if and only if $\sum_{p \in S} p^{1-\beta} < \infty$.

Proof. The Gibbs weight of the Hecke quotient sums one term per *left* coset $GL_2(\mathbb{Z})g$, equivalently per sublattice $L \subseteq \mathbb{Z}_p^2$, weighted by its index [1] — not one term per double coset. The number of sublattices of $\mathbb{Z}_p^2$ of index p^n is $\sigma(p^n) = 1 + p + \cdots + p^n$, and

$$\sum_{n \geq 0} \sigma(p^n) p^{-\beta n} = \sum_{n \geq 0} \frac{p^{n+1} - 1}{p - 1} p^{-\beta n} = (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1}.$$

Convergence of the Euler product is convergence of $\sum_p (p^{-\beta} + p^{1-\beta})$, dominated by the second term. $\square$

2. THE OBSTRUCTION SERIES AND ITS L -FUNCTION

Lemma 2.1 (Hilbert–Schmidt identity). *Let $f \in S_2(\Gamma_0(N))$ be a normalized cuspidal Hecke eigenform, $p \nmid N$, and $U_p = p^{-1/2} \rho_f(\text{Frob}_p)$, unitary with eigenvalues $e^{\pm i\theta_p}$ by Deligne’s theorem [2]. Then*

$$\|I_2 - U_p\|_{\text{HS}}^2 = 2(2 - \Re \text{Tr } U_p) = 4(1 - \cos \theta_p) = 8 \sin^2(\theta_p/2),$$

so that

$$(2.1) \quad \Xi_\beta(\pi_f, S) = 8 \sum_{p \in S} p^{-\beta} \sin^2(\theta_p/2).$$

Proof. Lemma 2.2 of Paper VI (its Trace form) with $n = 2$, together with $\text{Tr } U_p = a_p/\sqrt{p} = 2 \cos \theta_p$. $\square$

Theorem 2.2 (The L -function dictionary, corrected). *For $\beta > 1/2$,*

$$(2.2) \quad \Xi_\beta(\pi_f, S) = 4 \log \zeta_S(\beta) - 2 \Re \log L_S(\beta, \pi_f) + O(1),$$

with $L_S(s, \pi_f) = \prod_{p \in S} ((1 - e^{i\theta_p} p^{-s})(1 - e^{-i\theta_p} p^{-s}))^{-1}$ the analytically normalized standard L -function. No symmetric square factor occurs.

Proof. By (2.1), $\Xi_\beta = 4 \sum_p p^{-\beta} - 2 \sum_p p^{-\beta} \text{Tr } U_p$. The first sum is $\log \zeta_S(\beta) + O(1)$; the second is $\Re \log L_S(\beta, \pi_f) + O(1)$, the prime-power terms converging for $\beta > 1/2$. This is Theorem 4.1 of Paper VI at $n = 2$, where the general form is $2n \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \rho) + O(1)$. $\square$

Remark 2.3 (Why Sym^2 cannot appear). The proposed formula adds $-2 \Re \log L_S(\beta, \text{Sym}^2 \pi_f)$. That term cannot be present: $\|I - U\|_{\text{HS}}^2 = 2(2 - \Re \text{Tr } U)$ is *linear* in $\text{Tr } U_p$, whereas Sym^2 enters through $\text{Tr}(U_p^2) = \text{Tr}(U_p)^2 - 2 \det U_p$, a quadratic expression. Rankin–Selberg does govern the second moment: $\log L_S(s, \pi \times \pi^\vee) = \sum_p |\text{Tr } U_p|^2 p^{-s} + O(1)$, whose pole at $s = 1$ is the input to Sato–Tate. That is a different series from Ξ_β .

Corollary 2.4 (Divergence at $\beta = 1$). *For $S = P_{\mathbb{Q}}$ and any cuspidal eigenform f , $\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$.*

Proof. As $\beta \downarrow 1$, $\log \zeta(\beta) \rightarrow \infty$ while $L(\beta, \pi_f)$ has neither zero nor pole at $\beta = 1$ — Jacquet–Shalika non-vanishing on $\Re s = 1$ [4], and cuspidality excluding a pole — so $\Re \log L_S(\beta, \pi_f) = O(1)$. Apply (2.2). $\square$

3. THE GAP IS EXACTLY ONE

Definition 3.1. For S a set of primes let $\beta_c(S) = \inf\{\beta : \sum_{p \in S} p^{-\beta} < \infty\}$, so $0 \leq \beta_c(S) \leq 1$.

Theorem 3.2 (Disjoint ranges). *For every set S of primes and every automorphic representation π of GL_2 unramified outside S :*

- (1) $\Xi_\beta(\pi, S) < \infty$ for every $\beta > \beta_c(S)$, so no representation is an obstruction there; the obstruction can be nontrivial only for $\beta \leq \beta_c(S)$;
- (2) $\zeta_{GL_2, S}(\beta) < \infty$ for $\beta > 1 + \beta_c(S)$ and $= \infty$ for $\beta < 1 + \beta_c(S)$ — at the boundary $\beta = 1 + \beta_c(S)$ itself, finite exactly when $\sum_{p \in S} p^{-\beta_c(S)}$ converges — so Gibbs states of the GL_2 system exist only for $\beta \geq 1 + \beta_c(S)$, and only beyond it when the critical series diverges.

Consequently

$$(0, \beta_c(S)] \cap [1 + \beta_c(S), \infty) = \emptyset,$$

and the open interval $(\beta_c(S), 1 + \beta_c(S))$, of width exactly 1, lies in neither region. For $S = P_{\mathbb{Q}}$ the obstruction lives on $(0, 1]$ and the equilibrium states on $(2, \infty)$.

Proof. (1) By (2.1), $\Xi_\beta(\pi, S) \leq 8 \sum_{p \in S} p^{-\beta}$, finite for $\beta > \beta_c(S)$. (2) By Proposition 1.2, convergence is that of $\sum_{p \in S} p^{1-\beta} = \sum_{p \in S} p^{-(\beta-1)}$, which for $\beta - 1 > \beta_c(S)$ converges, for $\beta - 1 < \beta_c(S)$ diverges, and at $\beta - 1 = \beta_c(S)$ is the critical series itself. $\square$

Corollary 3.3 (The programme cannot close). *There is no temperature at which the localized GL_2 system possesses equilibrium states and the automorphic obstruction $\Xi_\beta(\pi, S)$ is capable of producing symmetry breaking. In particular the proposed uniqueness theorem on $(0, 1]$ concerns a range in which the GL_2 system has no Gibbs states, and Corollary 2.4, though true, has no KMS consequence for this system.*

Remark 3.4 (Numerical illustration). For $S = P_{\mathbb{Q}}$: $\sum_{p \leq 10^6} p^{-\beta}$ equals 4.79, 2.89, 1.99 at $\beta = 0.9, 1.0, 1.1$, the first two growing without bound and the third converging; while $\zeta(\beta)\zeta(\beta-1)$ is divergent at $\beta \leq 2$ and equals 8.33, 1.98, 1.30 at $\beta = 2.2, 3, 4$. The window $(1, 2]$ has neither.

Remark 3.5 (The shift is the multiplicity–norm mismatch). The displacement by 1 is the phenomenon isolated in Theorem 1.2 of Paper XII. There, for a maximal order in a central simple algebra of degree d , the number of right ideals of reduced norm $\mathfrak{p}$ is $d_{\mathfrak{p}} = (q^d - 1)/(q - 1) = 1 + q + \dots + q^{d-1}$, strictly larger than the norm $q = N\mathfrak{p}$; for $d = 2$ the excess is exactly the point at infinity of $\mathbb{P}^1$. Here the same count — $p + 1$ sublattices of index p in $\mathbb{Z}_p^2$ against norm p — produces the factor $(1 - p^{1-\beta})^{-1}$ in (1.1) and shifts the abscissa by 1. In Paper XII the excess emptied the boundary quotient; here it displaces the equilibrium range out of reach of the obstruction. It is one fact with two consequences.

4. SPARSE SETS FOR Δ

Theorem 4.1 (Anomalous sets exist). *Let $\Delta \in S_{12}(SL_2(\mathbb{Z}))$ with $\tau(p) = 2p^{11/2} \cos \theta_p$. Assuming the Sato–Tate equidistribution for Δ — a theorem of Barnet-Lamb, Geraghty, Harris and Taylor [3] — for every $\delta > 0$ the set $\{p : \theta_p \leq \delta\}$ has positive density. Choosing finite blocks $S_j \subseteq \{p : \theta_p \leq 2^{-j}\}$ with $1 \leq \sum_{p \in S_j} p^{-1} \leq 2$ and putting $S_\Delta = \bigsqcup_j S_j$ gives*

$$\sum_{p \in S_\Delta} p^{-1} = \infty, \quad \Xi_1(\pi_\Delta, S_\Delta) \leq \sum_j 2 \cdot 4^{-j} \cdot 2 < \infty.$$

Proof. $8 \sin^2(\theta/2) \leq 2\theta^2$, so the j -th block contributes at most $2 \cdot 4^{-j} \sum_{p \in S_j} p^{-1} \leq 4 \cdot 4^{-j}$. This is Theorem 5.3 of Paper VI with Δ in place of an elliptic curve. $\square$

Remark 4.2 (But not symmetry breaking). Theorem 4.1 produces a set on which the analytic obstruction is finite. It does not produce a broken phase, for two reasons already recorded. By Remark 5.4 of Paper VI an obstruction group is not a broken phase until an algebra realizes it; and by Theorem 3.2 the temperature $\beta = 1$ at which the construction operates is one at which this GL_2 system has no equilibrium states, $\beta_c(S_\Delta) = 1$ forcing Gibbs states only above 2. The residual symmetry subgroup $H_\Delta \subseteq SU(2)$ asked for is therefore not defined: there is no simplex on which it could act.

5. ASSESSMENT

Proposal	Status	Reference
$C(Y) \rtimes M_2(\mathbb{Z}_S)^+$	no: $ P^\times = \infty$	Prop. 1.1
$\ I_2 - U_p\ _{\text{HS}}^2 = 8 \sin^2(\theta_p/2)$	true	Lem. 2.1
Ξ_β involves Sym^2	false: spurious term	Rem. 2.3
$\Xi_\beta = 4 \log \zeta_S - 2\Re \log L_S + O(1)$	true	Thm. 2.2
$\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$	true, Jacquet–Shalika	Cor. 2.4
uniqueness on $(0, 1]$	vacuous: no KMS states there	Cor. 3.3
obstruction region	$(0, \beta_c(S)]$, $\beta_c \leq 1$	Thm. 3.2
equilibrium region	$(1 + \beta_c(S), \infty)$	Thm. 3.2
the two overlap	never; gap exactly 1	Thm. 3.2
sparse S_Δ with $\Xi_1 < \infty$	true, under Sato–Tate	Thm. 4.1
residual symmetry $H_\Delta \subseteq SU(2)$	not defined	Rem. 4.2

The arithmetic side of the proposal is sound and largely already proved in Paper VI: the Hilbert–Schmidt series, its L -function form once the spurious symmetric square is removed, the divergence at $\beta = 1$ by Jacquet–Shalika, and the construction of Sato–Tate anomalous sets. What fails is the junction with statistical mechanics, and it fails quantitatively rather than vaguely. The obstruction is a Dirichlet series in $p^{-\beta}$, with abscissa at most 1; the GL_2 partition function counts sublattices, whose local factor $(1 - p^{1-\beta})^{-1}$ moves the abscissa to $1 + \beta_c(S)$. The gap is exactly 1 and it is the same off-by-one of $\mathbb{P}^1$ that Paper XII showed to be universal.

Two ways out suggest themselves and neither is free. One could weight the dynamics differently, replacing $|\det g|$ by another multiplicative norm; but by Proposition 1.4 of Paper X norms factor through the abelianization, and on GL_2 orbits the abelianization is generated by $\det$, so any admissible norm is $|\det|^t$ and merely rescales β , leaving the ratio of the two abscissae fixed at $(1 + \beta_c)/\beta_c$ rather than closing the gap. Or one could keep the GL_1 normalization and attach the automorphic data as a coefficient system rather than as the acting semigroup — which is the route of §6 of Paper VI, where the Kakutani analysis applies verbatim and only the existence of a tractable expectation was left open. We would pursue the second.

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